ML-DSA.Sign(sk, M, ctx)

Generates an ML-DSA signature.

Input: Private key $sk \in \mathbb{B}^{32+32+64+32\cdot((\ell+k)\cdot\text{bitlen}\,(2\eta)+dk)}$, message $M \in \{0,1\}^*$, context string ctx (a byte string of 255 or fewer bytes).
Output: Signature $\sigma \in \mathbb{B}^{\lambda/4+\ell\cdot 32\cdot(1+\text{bitlen}\,(\gamma_1-1))+\omega+k}$.

if $|ctx| > 255$ **then**
  return $\perp$ ▷ return an error indication if the context string is too long
end if

$rnd \leftarrow \mathbb{B}^{32}$ ▷ for the optional deterministic variant, substitute $rnd \leftarrow \{0\}^{32}$
if $rnd =$ NULL **then**
  return $\perp$ ▷ return an error indication if random bit generation failed
end if

$M' \leftarrow$ BytesToBits(IntegerToBytes$(0,1)$ $\|$ IntegerToBytes$(|ctx|,1)$ $\|$ ctx) $\|$ M
$\sigma \leftarrow$ ML-DSA.Sign_internal(sk, M', rnd)
return σ